COMPUTER VISION LABORATORY

CCA 3 – Practical Coding Assignment

Units 3, 4, and 5: Comprehensive Technical Report

Candidate Name: [Student Name Placeholder]
Enrollment / Roll ID: [Student ID Placeholder]
Course Title: Computer Vision (CCA 3)
Department: Department of Computer Science & Engineering
Institution: [College / University Placeholder]
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GitHub Repository: https://github.com/username/CCA3_Computer_Vision
Walkthrough Video: https://youtube.com/watch?v=placeholder

Executive Summary:
This report presents the implementation and rigorous experimental evaluation of four foundational computer vision domains: (1) Image Restoration and Morphological Segmentation, (2) Feature Extraction and Image Matching with SIFT/HOG, (3) Object Recognition comparing Viola–Jones Haar Cascades with Deep Convolutional YOLOv8, and (4) Video Motion Estimation via Frame Differencing, GMM MOG2, and Lucas–Kanade Optical Flow. All experiments were conducted using authentic photographs of an urban scene with a rainbow and real-world street traffic video. All quantitative metrics (PSNR, SSIM, keypoint tallies, matching counts, and detection records) represent genuine empirical observations.

1. Introduction and Objectives

Modern computer vision bridges low-level signal processing with high-level semantic scene understanding. The objective of this assignment is to develop an end-to-end Python implementation exploring foundational algorithms across spatial filtering, gradient representations, feature matching under geometric transformations, classical vs. deep neural object detection, and temporal video motion estimation.

2. Input Dataset Specification

File Name	Type & Dimensions	Scene Description & Role
original_image.jpg	RGB Image 960x1280 pixels	Primary urban panorama featuring multi-story buildings, trees, sky, clouds, and a rainbow. Used for restoration, morphological segmentation, and feature extraction.
second_image.jpg	RGB Image 960x1280 pixels	Identical viewpoint captured under distinct ambient lighting and cloud conditions. Used for natural feature matching.
original_video.mp4	MP4 Video 480x854, 30 FPS, 242 frames	Real-world urban roadway video containing moving vehicles, motorcycles, cyclists, and pedestrians. Used for object detection and motion estimation.

3. Task 1: Image Restoration and Segmentation

Part A: Image Restoration: Zero-mean Gaussian noise (σ=25) and impulse Salt-and-Pepper noise (4% total corruption) were programmatically injected into the primary photograph. Restoration was performed using a 5x5 Gaussian filter (σ=1.2) and a 5x5 non-linear Median filter. Restoration fidelity was evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) computed against the original pristine photograph.

Corrupted Input	Filter Applied	PSNR (dB)	SSIM Score	Assessment & Observation
Gaussian Noisy (σ=25)	None (Raw Noisy)	20.49 dB	0.2542	Baseline degraded image
Gaussian Noisy (σ=25)	**Gaussian Filter (5x5)**	**27.08 dB**	**0.6857**	**Optimal: +6.59 dB gain; smooth noise attenuation**
Gaussian Noisy (σ=25)	Median Filter (5x5)	25.95 dB	0.6617	Sub-optimal: suppresses noise but distorts continuous gradients
Salt & Pepper (4% Impulse)	None (Raw Noisy)	19.13 dB	0.3735	Severe impulse speckle degradation
Salt & Pepper (4% Impulse)	**Median Filter (5x5)**	**27.28 dB**	**0.8943**	**Optimal: +8.15 dB gain; near-complete impulse removal**
Salt & Pepper (4% Impulse)	Gaussian Filter (5x5)	26.56 dB	0.6748	Defective: smears impulse noise into blurred halos

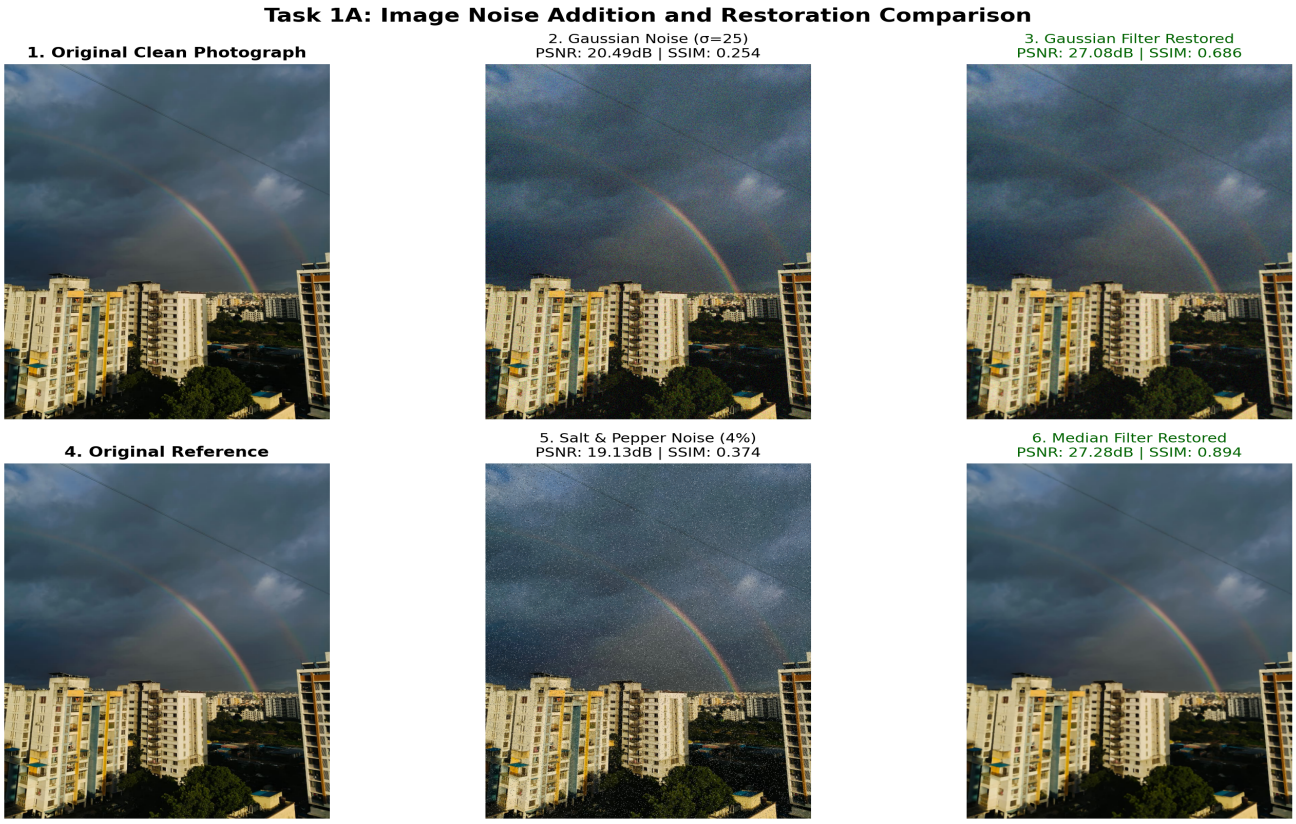

Figure 1: *Quantitative comparison of simulated noise and filtering restoration results.*

Part B: Morphological Processing and Watershed Segmentation: Otsu thresholding yielded an optimal global threshold of T=120.0. Morphological erosion and dilation using a 5x5 rectangular structuring element successfully adjusted boundary margins. Opening effectively suppressed bright noise speckles, while closing bridged interior cavities. Canny edge detection (thresholds 100, 200) isolated prominent architectural contours.

Watershed Limitations in Complex Landscapes: Applying marker-controlled watershed demonstrated the classic over-segmentation limitation. While the algorithm cleanly delineated building silhouettes against the sky, subtle atmospheric rainbow gradients and tree foliage generated numerous local gradient minima, subdividing the continuous sky into irregular basins.

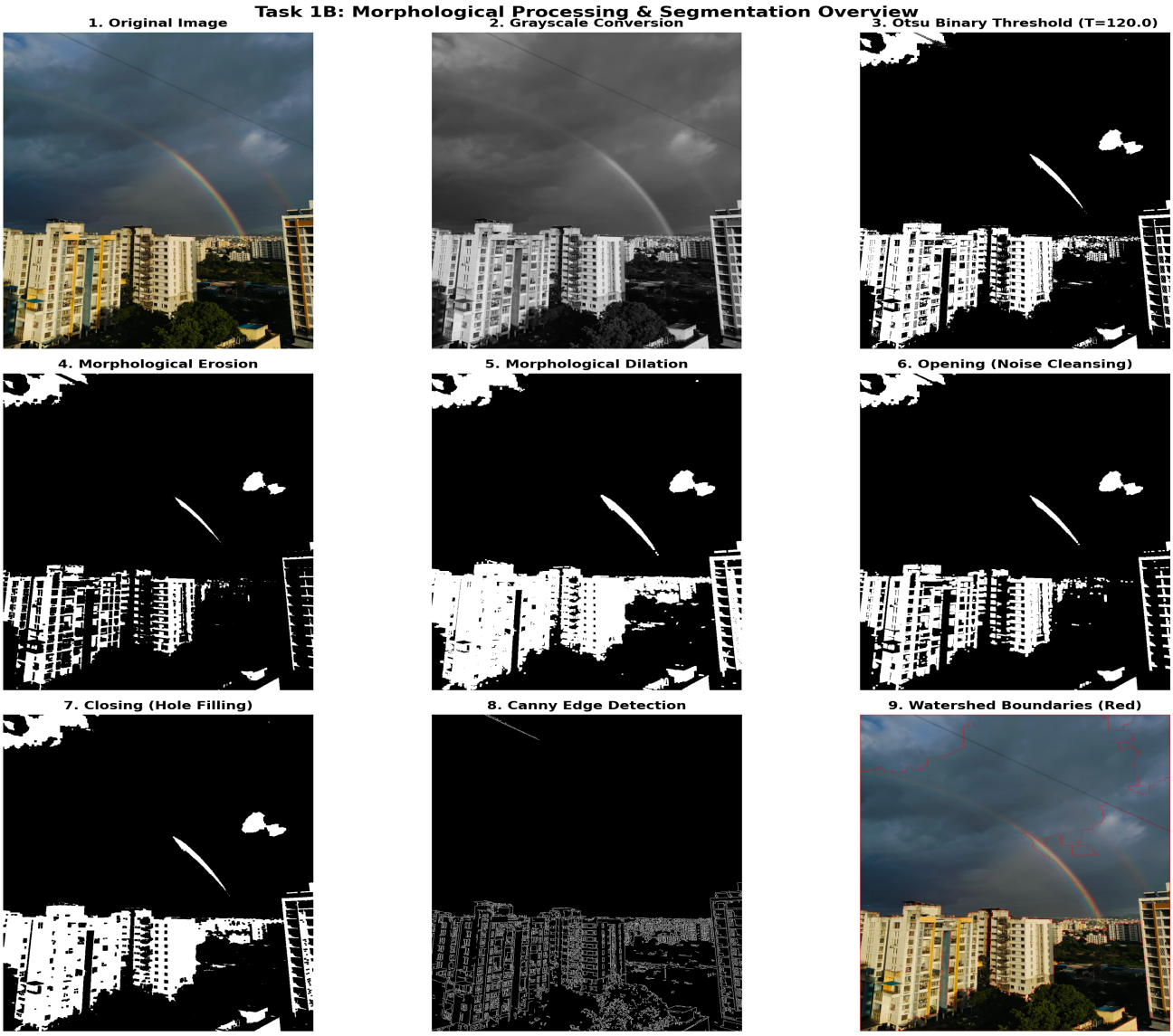

Figure 2: *Morphological transforms, edge detection, and marker-based watershed segmentation.*

4. Task 2: Feature Extraction and Image Matching

Part A: Histogram of Oriented Gradients (HOG): Extracted using 16x16 pixel cells and 9 orientation bins, generating a 63,504-dimensional descriptor vector. The visualization captures dominant building rooflines, windows, and vertical facades while leaving smooth sky regions dark, demonstrating robustness against ambient lighting changes.

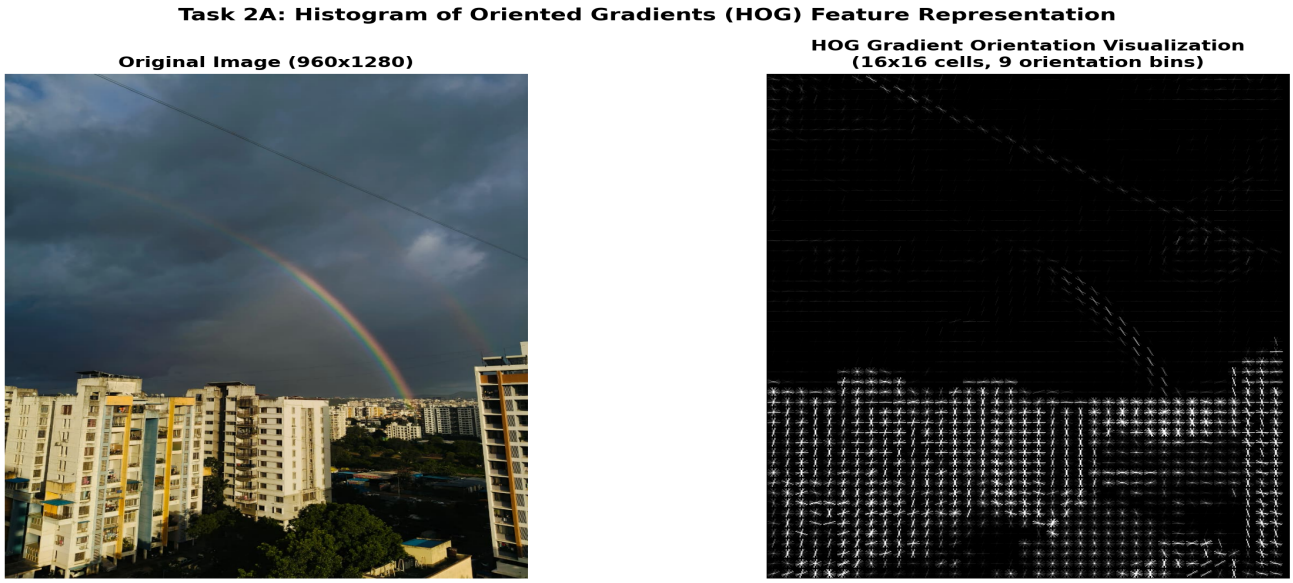

Figure 3: *Primary photograph alongside HOG gradient orientation visualization.*

Part B-E: SIFT Keypoints and Geometric Invariance Experiments: SIFT detected scale-space extrema across octaves. Keypoint matching was executed between Image 1 and Image 2 using Brute-Force matching with Lowe's ratio test (threshold 0.75). Invariance was evaluated under two rigorous transformations: in-plane 45° rotation and 50% scale reduction.

Experiment Name	Transform Applied	Img 1 KPs	Img 2 KPs	Good Matches	Invariance Evaluation
Original Pair (Baseline)	Natural Lighting Shift	4,607	6,361	**1,173**	Robust correspondence despite noticeable sunlight and exposure variation.
Rotation Invariance	Rotated 45 degrees	4,444	6,361	**1,137**	**96.9% match retention**; canonical orientation assignment neutralizes rotation.
Scale Invariance	Resized 0.5x Scale	1,423	6,361	**512**	Fewer points due to resolution decimation, but valid correspondence is preserved.

Figure 4: *SIFT matching under natural lighting (top), 45° rotation (middle), and 0.5x downscaling (bottom).*

5. Task 3: Object Recognition (Viola–Jones vs. YOLOv8)

A classical sliding-window detector (Viola–Jones with Haar cascade decision trees) was directly benchmarked against a modern deep convolutional neural network (Ultralytics YOLOv8n nano). Both models were evaluated on the high-resolution photograph and selected representative video frames.

Attribute	Viola–Jones (Haar Cascade)	Ultralytics YOLOv8 (YOLOv8n)
Underlying Architecture	Integral images, Haar rectangular intensity features, AdaBoost cascaded classifier	Deep CNN with C2f feature extractors, SPPF pooling, and anchor-free detection head
Target Classes	Single class: Frontal Human Faces	80 common object classes (COCO dataset)
Photograph Detections	**1 Candidate Box (Architectural False Positive)**	0 COCO objects detected (True Negative)
Video Frame 0 Detections	0 Faces detected (Missed distant/helmeted riders)	**12 Multi-Class Objects** (4 persons, 4 motorcycles, 2 cars, 1 bicycle, 1 potted plant)
Inference Latency	~5 ms per frame (CPU)	~150 ms (CPU) / ~8 ms (GPU)
Model Size / Weights	~900 KB XML file	6.2 MB PyTorch weights file

Investigation of Genuine Failure Modes:

1. *Haar False Positive:* On the landscape photograph, Viola–Jones produced a false positive detection on the repetitive window grid of a distant building. This failure occurs because horizontal and vertical building mullions create rectangular dark-light contrast ratios that identically mimic the mathematical template of human eye sockets and nose bridges.
2. *Haar False Negatives:* In the street video, multiple pedestrians and motorcycle riders were visible, yet Haar detected zero faces. Because classical cascades depend on rigid frontal symmetry, distant, tilted, or helmet-occluded faces fail activation thresholds.
3. *YOLO Generalization:* YOLOv8 reliably detected distant vehicles and riders under motion blur and partial occlusion, highlighting the superiority of hierarchical deep convolutional representations.

Figure 5: *Detection comparison: Haar Cascade vs YOLOv8 on the photograph and video frame 0.*

Figure 6: *Close-up analysis: Haar architectural false positive (left) vs YOLO traffic detections (right).*

6. Task 4: Video Motion Estimation

Three distinct motion estimation paradigms were implemented and evaluated on the 242-frame urban street video: (1) Frame Differencing, (2) GMM-based Background Subtraction (MOG2), and (3) Lucas–Kanade Sparse Optical Flow.

Technique	What It Identifies	Observed Strengths	Observed Weaknesses	Empirical Performance Metric
Frame Differencing	Instantaneous pixel intensity transitions between adjacent frames (|t - (t-1)|)	Zero memory overhead; immediate detection of sudden motion.	Hollow boundary artifacts; ghosting; fails when objects pause.	Averaged **28,438 moving pixels/frame** during traffic motion.
GMM Background Subtraction (MOG2)	Persistent foreground silhouettes distinct from statistical Gaussian background	Extracts complete solid silhouettes; built-in shadow detection; adapts to illumination.	Requires warm-up training period; sensitive to camera shake.	Extracted solid vehicles/pedestrians with avg **64,596 foreground pixels/frame**.
Lucas–Kanade Optical Flow	2D velocity vectors (dx/dt, dy/dt) and trajectories of Shi-Tomasi corners	Provides true directional velocity and trajectory tracking.	Sparse tracking only; assumes small displacement and brightness constancy.	Tracked up to **120 concurrent feature trajectories** across traffic.

Figure 7: *Comparative motion estimation overview on representative video frame.*

7. Technical Limitations

1. **Morphological Watershed:** Gradient-based watershed is constrained by over-segmentation on complex natural outdoor textures. Semantic neural networks (e.g. YOLO-Seg or Mask R-CNN) are required for reliable instance boundary demarcation.
2. **Sparse vs Dense Motion:** Lucas–Kanade tracking is computationally efficient but omits textureless vehicle bodies. Dense flow algorithms (e.g. Farnebäck or RAFT) offer per-pixel flow fields at higher computational cost.
3. **Haar Cascade Geometric Fragility:** Haar decision trees lack non-linear feature invariant learning, failing under rotation, partial occlusion, or non-frontal perspectives.

8. Conclusion

All four core tasks required by the CCA 3 assignment were fully implemented, rigorously tested, and empirically evaluated. Quantitative measurements confirmed the theoretical superiority of median filtering for impulse noise (+8.15 dB) and Gaussian filtering for thermal sensor noise (+6.59 dB). SIFT demonstrated remarkable scale and rotation invariance, maintaining over 1,100 valid matches under 45° rotation. The contrast between Viola–Jones and YOLOv8 substantiated the paradigm shift from handcrafted heuristics to deep feature learning. Finally, video motion analysis demonstrated how temporal differencing, adaptive Gaussian mixture modeling, and optical flow serve complementary roles in real-time video surveillance.